

■ SeriesData

`SeriesData[x, x0, {a0, a1, ...}, nmin, nmax, den]` represents a power series in the variable x about the point x_0 . The a_i are the coefficients in the power series. The powers of $(x-x_0)$ that appear are $nmin/den, (nmin+1)/den, \dots, nmax/den$.

`SeriesData` objects are generated by `Series`. ■ `SeriesData` objects are printed as sums of the coefficients a_i , multiplied by powers of $x - x_0$. A `SeriesData` object representing a power series is printed with `0 [x]^p` added, to represent omitted higher-order terms. ■ When you apply certain mathematical operations to `SeriesData` objects, new `SeriesData` objects truncated to the appropriate order are produced. ■ The operations you can perform on `SeriesData` objects include arithmetic ones, mathematical functions with built-in derivatives, and integration and differentiation. ■ `Normal[expr]` converts a `SeriesData` object into a normal expression, truncating omitted higher-order terms. ■ If the variable in a `SeriesData` object is itself a `SeriesData` object, then the composition of the `SeriesData` objects is computed. Substituting one series into another series with the same expansion parameter therefore automatically leads to composition of the series. Composition is only possible if the first term of the inner series involves a positive power of the variable. ■ `InverseSeries` can be applied to `SeriesData` objects to give series for inverse functions. ■ See page 430.